

Changes in Modularity with the Random Incorporation of a Species to Plant-Pollinator Networks

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1 Introduction

A community with an ecological/evolutionary mutualism can be represented as a bipartite graph, where nodes are species (or, more rarely, individuals), connected by an edge if there is a known interaction between them. These interactions can be observed in *in situ* field conditions or by other means. These mutualistic networks have been observed to have modular structure, where nodes can be partitioned into subsets within which connectivity is high but among which connectivity is low [16] [17]. Modules are often formed by hubs of generalist plant species that attract many pollinators, with few connector species that link modules to each other [7]. This modular structure has been associated in other network types to properties such as persistence [19] and stability [14]. Modular structure in plant-pollinator networks specifically has been linked to lower persistence [2] and may be the signature of coevolutionary units [7].

When exotic or invasive species enter a plant-pollinator network, the modular network structure may change. Studies of invasive species have found that invaded networks tend to be less modular with fewer modules than uninvaded networks [22] [23]. These species are able to directly integrate into the network [5] becoming network hubs [23]. These invasions may decimate connections among native species [25] as interactions shift from native to invasive species [1]. These changes to the topology of the network could have evolutionary or ecological implications at the species or community level.

Studies thus far of how mutualistic network properties change with invasion have studied networks post-hoc, considering how modularity or other measures differ in networks that have varying proportions of invasive species [1] [4] [22] [23], or at one point in an invasion process [18]. This project simulates random incorporations of a single invasive species into a network, recording the change in modularity and differences in the modular memberships. This study will only investigate the incorporation of alien plants into the community, and will only consider a single nondescript invasive plant species, so as to observe changes with only relatively small changes in the network.

2 Methods

A set of three plant pollinator networks were collected from the R package `bipartite` 2.05 [6]. These networks were stored in the form of weighted incidence matrices. In order to perform clustering, these matrices were converted into unweighted `igraph` graph objects. Modularity was calculated using the `igraph cluster_fast_greedy` routine, with a resulting object containing module membership for each node and a modularity score. The modularity score of a partition into k modules is

$$M = \sum_{s=1}^k \left(\frac{l_s}{L} - \left(\frac{d_s}{2L} \right)^2 \right)$$

Where l_s is the number of links (edges) within the s th module, d_s is the sum of degrees within the s th module, and L is the total number of links in the network. The modularity score of a network is the maximum M achieved over all possible partitions [9]. The modules created in the partition which achieves this maximum are

called the modules of the network. Because of the complexity involved in searching over all possible partitions, the `cluster_fast_greedy` routine uses a greedy algorithm rather than an integer linear program.

I simulate five different ways that an invasive plant species may integrate into a network.

1. Random assignment of a pre-determined number of links. In these trials, the expected number of links to be added is uniformly sampled between 0 and $\frac{1}{6}$ of the total number of pollinators in the network; the links are then randomly distributed among pollinators. This allows data to be collected for plant species on a spectrum of specialist (few links added) to super-generalist (many links added). Invasive plants have been observed to range from relatively poorly-connected [15] to super-generalist [1] [22].
2. Random integration of links within a predetermined module. This assigns links to species within the module with probability p_{in} , and links to species not inside the module with probability p_{out} , where $p_{in} > p_{out}$. This simulates the integration of a vertex into a single module, and follows the model of random modular graph generation [9]. In this study, $p_{in} = 0.4$, $p_{out} = 0.04$. However, this is not necessarily biologically realistic [15], as invasive species often assume hub-connecting roles [5].
3. Random assignment of links to pollinators with probability proportional to observed number of interactions. This was done using the formula

$$\Pr[\text{link from exotic to species } i] = 0.75 \sqrt{\frac{w_i}{\max_j (w_j)}}$$

Where w_j is the total number of observed interactions (i.e., sum of weights of edges incident to node j in a weighted interaction network) of the i th species. This normalization ensures that the most likely pollinator to interact with the invasive plant only has a 75% chance of realizing this interaction (75% is an arbitrarily chosen number). It is logical and intuitive to claim that a species that is observed more often pollinating during data collection is more likely to pollinate a new species than not. The roots of interactions are used rather than the number of interactions because of the high variance in the interaction distribution, which biases interactions with high-interaction pollinators over low-interaction ones. Using the roots of edge weights reduces the variance and assigns edges more evenly.

4. Random assignment of links to pollinators with probability proportional to pollinator degree. This is done using the formula

$$\Pr[\text{link from exotic to species } i] = 0.45 \sqrt{\frac{d_i}{\max_j (d_j)}}$$

Where d_i is the degree of the i th pollinator. The scalar 0.45 was chosen to create a number of links comparable to those generated by scheme 3. Studies of webs with invasive species show that generalist pollinators are more likely to interact with invasive plants than non-generalist pollinators [15] [18] [20] [23].

5. Random assignment of links to pollinators with probability inversely proportional to pollinator degree. This is done using the formula

$$\Pr[\text{link from exotic to species } i] = 0.135 \frac{1/d_i}{\max_j (1/d_j)}$$

This is equivalent to

$$\frac{0.135 \min_j (d_j)}{d_i}$$

The scalar 0.135 was chosen to create a number of links comparable with those generated by schemes 3 and 4. It has been demonstrated that in some invaded pollinator networks, invasive plants interact with specialist species more often than expected [20]. However, this is not a universal pattern [15].

Edges are randomly assigned in 500 independent trials for each scheme. After edges were assigned, the difference between the modularity score of the new graph and the score of the original network (ΔM) was recorded. Additionally, the following partition distances is used (modified by Karrer et al. (2008) from van Dongen (2000)):

$$D(C, C') = 1 - \frac{1}{2n} \left(\sum_{k=1}^K \max_{k'} (|C_k \cup C'_{k'}|) + \sum_{k'=1}^{K'} \max_k (|C_k \cup C'_{k'}|) \right)$$

$D(C, C')$ is a normalized distance between two clusters C and C' , which respectively classify n objects into K and K' clusters. The two summands are the row- and column-maxima of a $K \times K'$ confusion matrix. $D(C, C')$ can be interpreted as the proportion of nodes that move between clusters when the new partition occurs; a score of 0 corresponds to identical clusters.

Plant invasions are marked by a shift in interactions from native plants to the exotic species, generally keeping the connectivity of invaded and non-invaded networks similar [1] [18] [20] [24]. To simulate this redistribution in links, for each new partner of the invasive plant, the links in the partner's original set of interactions with native plants are deleted with probability p_r using Monte Carlo simulation. For this study, a gradient of edge deletions is created by setting $p_r = 0, 0.10, 0.25$, and 0.50 .

These trials are run at first on a dataset from Kato et al., (1990) (henceforth known as Kato), which is included in `bipartite`. This dataset has connectivity 94 plants, 673 pollinators, modularity 0.611 and 3.8% of potential links realized. To compare trends across networks, I will also use datasets from Junker et al., (2013) (henceforth, Junker) and Kevan (1970) (henceforth, Kevan). Junker has 56 plants and 257 pollinators, modularity 0.515, with 7.9% of links realized. Kevan has 30 plants and 114 pollinators, with modularity 0.413 and 18.2% of links realized.

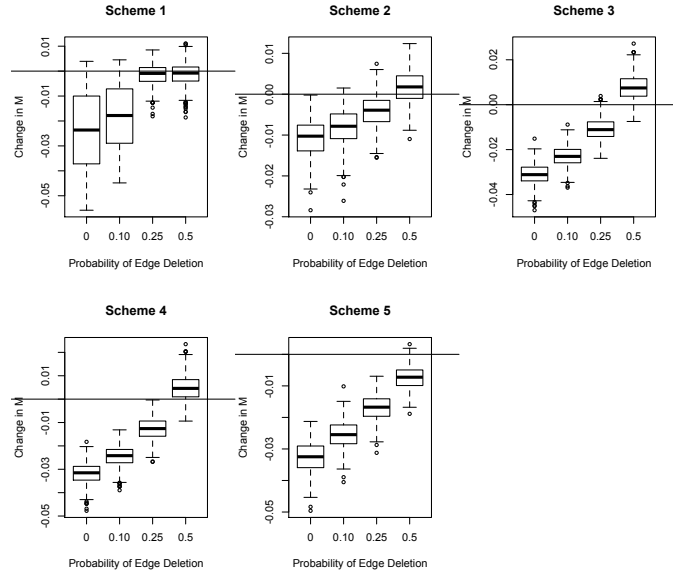
3 Results

Figure 1 contains box plots of modularity score changes and partition distances for all five rewiring schemes with varying rewiring probabilities in the Kato dataset. In all five schemes, modularity decreases in most trials, although the median decrease in modularity increases with increasing p_r . In the Kato network, modularity decreases tend to be larger in scheme 5 than in schemes 3 and 4. Modularity distance D' has a median close to 0.2 in all schemes and for all probabilities of edge deletion.

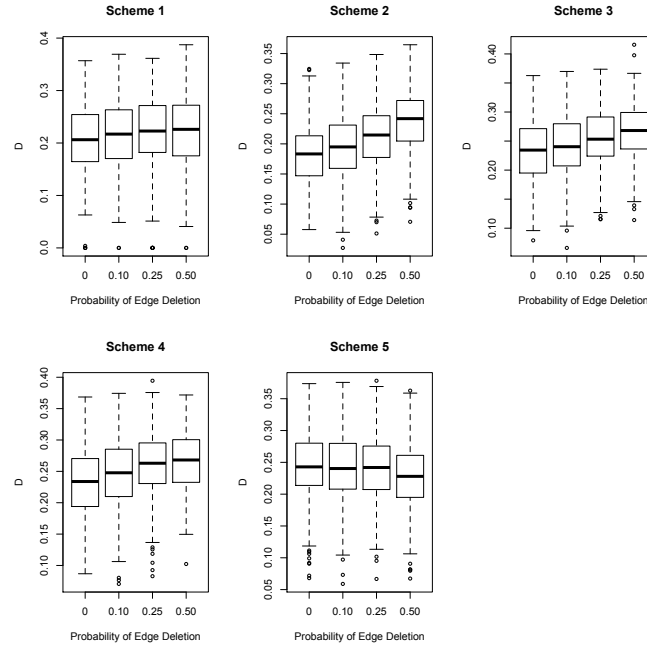
Regressing ΔM on the number of edges in the Kato dataset yielded significant relationships when $p_r = 0$. For $p_r = 0$, each additional interaction caused a decrease in modularity of between $3e^{-4}$ and $5e^{-4}$. In scheme 1, this relationship was able to explain 97% of variance in ΔM , but only about 65% of variance in schemes 3 – 5 and 30% of variance in scheme 2. With increasing p_r , regression coefficients stayed roughly within one order of magnitude, but the R^2 of these regressions decreased with increasing p_r , falling below 0.10 for $p_r = 0.5$. The effects of invasive species degree on modularity score remained significant at the $\alpha = 0.01$ level for all of these regressions except for one (scheme 2, $p_r = 0.5$), but was able to explain very little of the variation introduced by edge deletion. A table of regression coefficients and R^2 values can be found in Table 1.

Regressing D on the number of edges in the Kato dataset yielded some significant relationships (for schemes 1, 3 and 4) when $p_r = 0$. However, these relationships explained very little of the variation in D . Because of this poor explanatory power, regression analysis was not performed on D for other values of p_r or on other datasets. Summary of these regressions can be found in Table 2.

Histograms of ΔM and D in all three datasets with $p_r = 0$ can be found in Figure 2. With decreasing network size and modularity and increasing connectivity, the decrease in modularity becomes smaller (medians of ΔM move closer to zero). Also, the relative orderings of ΔM in schemes 3, 4 and 5 show no consistency moving among networks. Partition distances D are similar in Kato (range of sample medians: (0.183, 0.248)) and Junker (median range: (0.180, 0.236)), but smaller in the Kevan dataset (median range (0.057, 0.125)).

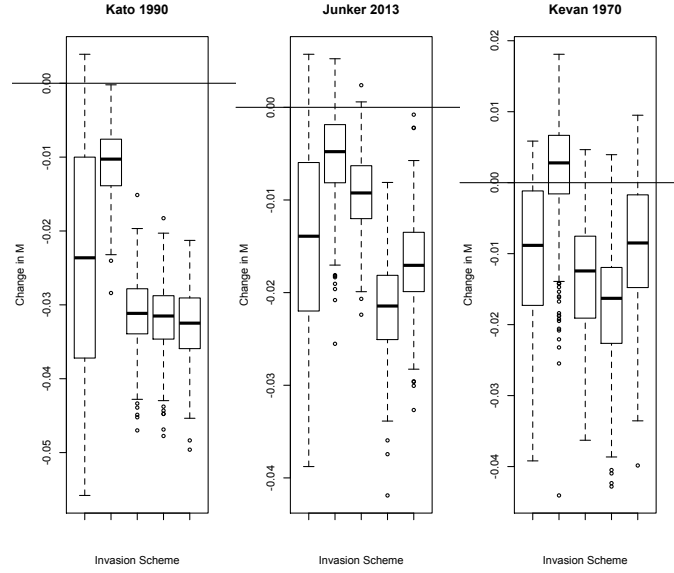


(a) Changes in modularity score. Solid line corresponds to $\Delta M = 0$ (no change).

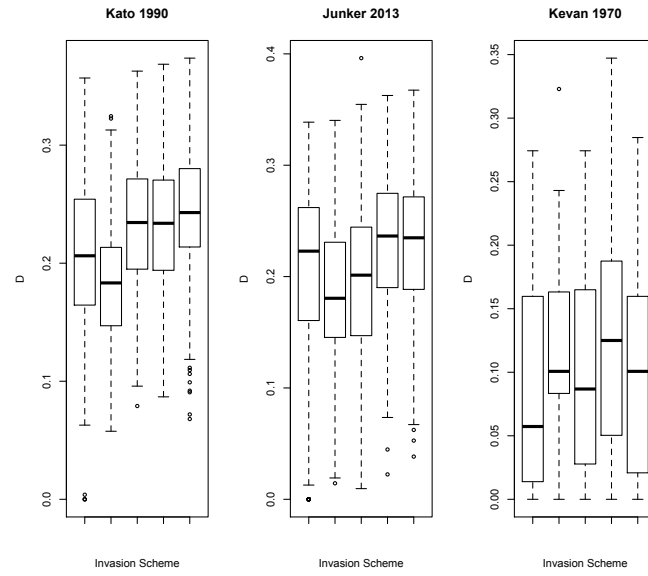


(b) Distances between partitions.

Figure 1: Results for Kato dataset with varying probabilities of edge deletion (respectively, from right to left, $p_r = 0, 0.10, 0.25, 0.50$). In all box plots, $n = 500$, and hairs are 1.5 times length of the IQR.



(a) Histograms of change in modularity. Solid line corresponds to $\Delta M = 0$.



(b) Histograms of partition distance D .

Figure 2: Histograms of ΔM and D for five schemes in Kato, Junker and Kevan networks, respectively. In all trials, edge deletion probability is $p_r = 0$.

	$p = 0$		$p = 0.10$		$p = 0.25$		$p = 0.50$	
Scheme	b_1	R^2	b_1	R^2	b_1	R^2	b_1	R^2
1	$-4.6e^{-4}$	0.97	$-3.7e^{-4}$	0.94	$-2.4e^{-4}$	0.82	$-6.4e^{-5}$	0.22
2	$-3.8e^{-4}$	0.30	$-3.0e^{-4}$	0.19	$-2.1e^{-4}$	0.10	$-3.2e^{-5}$	0.00
3	$-4.7e^{-4}$	0.65	$-4.4e^{-4}$	0.51	$-3.9e^{-4}$	0.31	$-2.1e^{-4}$	0.05
4	$-4.7e^{-4}$	0.66	$-3.9e^{-4}$	0.43	$-3.4e^{-4}$	0.25	$-2.0e^{-4}$	0.05
5	$-5.1e^{-4}$	0.69	$-4.5e^{-4}$	0.60	$-3.3e^{-4}$	0.35	$-1.3e^{-4}$	0.05

Table 1: Summaries of regressions of modularity change against edges added in Kato dataset, with various probabilities of edge deletion.

Scheme	b_1	p	R^2
1	$+9.93 * 10^{-4}$	0.000	0.269
2	$-2.21 * 10^{-5}$	0.946	0.000
3	$+8.44 * 10^{-4}$	0.003	0.015
4	$+6.50 * 10^{-4}$	0.028	0.009
5	$+2.81 * 10^{-4}$	0.352	0.001

Table 2: Summaries of regressions of partition distance against edges added in Kato dataset.

A binomial test on ΔM with null hypothesis $p = 0.5$ can test whether or not modularity changes have a specific direction (e.g., whether or not there is a greater probability of increase in modularity rather than decrease). These tests were performed on all three datasets for all levels of p_r . Results and 95% confidence intervals can be found in Table 3. Generally, these tests show that for low or zero- p_r , modularity decreases, although this effect decreases with increasing p_r . In the larger networks (Kato and Junker), when p_r increases to 0.5, the binomial test shows a significant deviation from $p = 0.5$, but the estimates of p suggest an increase in modularity rather than a decrease. In the smaller Kevan network, when p_r is low, modularity decreases, but the estimates of p are higher than in the Kato and Junker networks (i.e., there is a slightly higher probability of modularity increasing). Yet, when p_r increases to 0.5, binomial tests fail to detect a direction in the shift in ΔM .

4 Discussion

In trials where a node is added to a network with randomly assigned interactions to pre-existing nodes, modularity decreases if edges between pre-existing nodes are left intact. However, the decreases in modularity become smaller as edges between pre-existing nodes are deleted at random. The way that edges incident to the new node are added may affect how much change in the modularity occurs, although a comparison across networks shows does not reveal an immediately obvious trend or one-size-fits-all rule for how the edge assignment affects modularity. One exception is that decreases in modularity are smaller (or modularity may even increase) when purposefully incorporating the new node into a specific module. Ecologically, this suggests that most ways of randomly incorporating an invasive species into a plant-pollinator community will cause the network to become less modular. These decreases in modularity are consistent with prior research [2]. This is especially likely to occur if new partners of the invasive species do not abandon their native partners.

A clear relationship can be found where increasing generalization of the invasive species causes larger decreases, but only in cases where pollinators do not abandon their native hosts, a case that does not seem ecologically relevant or likely [1] [24]. This relationship becomes weaker and less significant as pollinator partners sever ties with native plants. Thus, although it has been demonstrated that invasive plants tend to be [1] [2] (but are not always [15] [22] [23]) generalists, this study can not conclude with certainty that super-generalist pollinators are more likely to eliminate modular structure.

Although only three networks were analyzed, trends noted above generally persisted across networks. These networks fell on a gradient of modularity, size, and connectivity. The larger, less connected, more modular networks

		$p_r = 0.00$		$p_r = 0.10$		$p_r = 0.25$		$p_r = 0.50$	
Data	Scheme	p	CI	p	CI	p	CI	p	CI
Kato	1	.000	(.005, .094)	.000	(.049, .096)	.000	(.363, .450)	.005	(.378, .467)
	2	.000	(.000, .007)	.000	(.006, .029)	.000	(.118, .182)	.000	(.628, .711)
	3	.000	(.000, .007)	.000	(.000, .007)	.000	(.012, .042)	.000	(.888, .939)
	4	.000	(.000, .007)	.000	(.000, .007)	.000	(.000, .007)	.000	(.781, .851)
	5	.000	(.000, .007)	.000	(.000, .007)	.000	(.000, .007)	.000	(.013, .044)
Junker	1	.000	(.031, .071)	.000	(.062, .114)	.000	(.140, .210)	.873	(.451, .541)
	2	.000	(.079, .134)	.000	(.177, .250)	.081	(.416, .505)	.000	(.712, .789)
	3	.000	(.002, .020)	.000	(.070, .123)	.623	(.443, .553)	.000	(.915, .959)
	4	.000	(.000, .007)	.000	(.000, .007)	.000	(.098, .154)	.000	(.846, .905)
	5	.000	(.000, .007)	.000	(.000, .014)	.000	(.010, .036)	.000	(.096, .156)
Kevan	1	.000	(.118, .182)	.000	(.193, .269)	.000	(.302, .387)	.447	(.437, .526)
	2	.000	(.624, .709)	.000	(.604, .690)	.000	(.589, .684)	.000	(.739, .814)
	3	.000	(.011, .039)	.000	(.056, .105)	.000	(.237, .327)	.983	(.451, .541)
	4	.000	(.002, .020)	.000	(.034, .075)	.000	(.157, .227)	.447	(.437, .537)
	5	.000	(.129, .195)	.000	(.205, .282)	.000	(.353, .441)	.227	(.483, .572)

Table 3: p values and 95% confidence intervals for two-sided binomial tests ($H_0: p = 0.5$) with $n = 500$. In this test, a success is modularity increasing – i.e., high p means higher probability of modularity increasing.

tended to have a slightly larger range in outcomes – invasions without cascading link-severing created less-modular networks, and cases with high link-severing caused some cases where modularity increased. In the smaller, more connected, less modular network, this range of outcomes was somewhat compressed. Invasions without link-severing still led to less modular networks, although there were more cases of modularity increasing. Likewise, invasions with high link-severing did not create networks that were consistently more positive or negative. Additionally, while whether or not an invasive species attracted specialist or generalist pollinators seemed to affect how large the change in modularity score was, there was no clear trend that persisted across networks. In the Kato and Junker networks, an invasive plant that favored generalist or high-interaction pollinators tended to cause a higher modularity score than an invasive plant that favored specialists, but this trend seemed to be reversed in the Kevan dataset. Whether or not these differences were due to the degree distributions, initial modularity, size or connectivity of the network is unclear – analysis of more networks with a variety of architectures would be necessary.

Modularity is a property known to scale with the square root of the number of links in the network [8] [9], and thus the connectivity as well. It is worth noting that this may be able to explain some of the general trends observed in this study. Adding the invasive species to the community and introducing interactions with native species without removing interactions between native species increases the number of edges, which one would expect to increase modularity. The random edge deletion could counteract the number of added edges and may return the overall number of edges to a number near the number of edges in the original network, which could create only a small difference in the modularity score. In cases where no edges were deleted, the number of edges added could explain over 95% of the variance in modularity when edges were randomly assigned, but less than 70% of variance when topological information about the nodes was used to assign edges. This suggests that the ways that edges are assigned (i.e., how the invasive species incorporates into the community) does have at least some effect on the modularity. If this research is to be continued, the first step should be to investigate the relationship between the change in the size of the edge set on the change in the modularity score. This could easily be done with a simple linear regression.

In the larger datasets studied, the differences in the composition of the communities was larger than expected and did not depend on the way the invader incorporated itself into the community or the invader degree. In at least half of the simulated invasions, at least 20% of species moved between modules. In the smallest, least modular network, the partitions were more similar to each other after invasion (with a median of roughly 10% of species moving between modules). Assuming that this is not caused by the routine used to calculate modules, this result is counterintuitive. One would expect that a network that is more modular to have modules that are more robust to small changes in the

network. However, these trials had the opposite result – the least modular network showed less change in modularity than the more modular networks.

Higher modularity tends to decrease network persistence [21]. An invasion stabilizing a network could make the network over-reliant on the invasive species [1] [5], and the network may collapse if the invasive species is removed [25]. This finding suggests that in most invasions, modularity is likely to decrease. Whether or not extant plant-pollinator networks are highly susceptible to invasion would require a comparison with random graphs, but these results suggest that most invasions will cause slight decreases in modularity and thus easy integration of invasive species. However, if the invaders are so attractive that they can cause pollinators to abandon their original partners and rely more on them, they may increase modularity and decrease overall stability, perhaps by making these invasive pollinators over-reliant on them. Because studies of stability and robustness of the network rely on other network properties such as nestedness [3] [21], it would be helpful to do similar analyses on network-wide properties and their robustness to simulated invasions.

The fact that modular partitions differ by 20% after relatively small changes to the network suggests that invasions have serious potential to change the makeup of individual modules. If modules are viewed as small coevolutionary units [7], then invasions can cause rapid changes in the evolutionary trajectory of these mutualistic networks by shuffling partners. However, this gulf in the similarity of partitions can be interpreted another way. Another metric to assess the quality of modular partitions is to study their robustness to small perturbations in the network [11]. Partitions that change with small perturbations can be viewed as less reliable or meaningful. If the modules that are created change drastically with the addition of one species, then perhaps those modules are not as biologically relevant or significant as originally thought. These simulations pose a question about how robust and meaningful these modules truly are. That question can be answered with a more systematic way to assess how module composition changes with changes in a network.

5 Conclusion

This study simulated the introduction of a nondescript invasive plant to historic plant-pollinator networks and assessed how the modularity changed under various conditions. In most cases, modularity decreased. However, the decrease in modularity was smaller, and modularity even increased, when pollinator partners of the invasive plant abandoned their original partners in response to the invasion. This trend tended to persist across networks, although the effects were less stated in a smaller, less modular network. The modular partitions showed higher change than expected, regardless of how the species is connected to the rest of the network. This suggests that most additions to plant-pollinator networks may lower modularity, and shift the composition of the modules. However, the change in network composition may suggest that the modules themselves are not as biologically relevant as originally thought.

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